CENTRAL BANK OF KENYA

Monetary Policy Committee | Policy Analysis Unit

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WHITE PAPER

Understanding Kenya's Interest Rates and Inflation:

What the Numbers Say and What It Means for Every Kenyan

THIS PAPER IS ADDRESSED TO:

H.E. The Governor, Central Bank of Kenya

The Chairperson & Members, Monetary Policy Committee (MPC)

The Cabinet Secretary, National Treasury & Economic Planning

The People of Kenya — Wananchi wa Kenya

Kila Mkenya — mtoto, mkulima, mfanyabiashara, mwanafunzi, dereva, muuzaji

LEAD SCIENTIST	UNIT	DATE	STATUS
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A NOTE TO EVERY READER BEFORE WE BEGIN

This paper is for everyone — not just economists.

You do not need to understand finance, economics, or mathematics to read this paper. We have written it the way a trusted friend would explain it over a cup of chai. Every technical idea is explained in plain English. Every number is given a real-life meaning. Because inflation and interest rates affect every Kenyan — whether you are a governor, a mama mboga, a farmer in Kisumu, a student in Nakuru, or a boda boda rider in Nairobi.

Kumbuka (Remember): Hii ni karatasi ya elimu tu.

This is a simulated document created for educational purposes. It is not an official publication of the Central Bank of Kenya. The analysis is based on real historical data but the paper itself is a learning tool built to demonstrate how central bank research works.

In this paper we walk through what 53 years of data tells us about how Kenya's interest rates and inflation have interacted — and what our deep statistical models reveal about how monetary policy actually works in practice.

1. WHAT ARE INTEREST RATES AND INFLATION? — MAANA HALISI

Before we look at any data or charts, let us make sure we all understand the two main ideas in this paper.

INFLATION — Bei za Bidhaa Kupanda

Inflation is the rate at which prices of everyday goods go up. If a kilo of unga cost KES 100 last year and costs KES 110 this year, inflation is 10%. When inflation is high, your money buys less than it used to. A salary of KES 30,000 feels like KES 27,000 if inflation is 10%. Inflation is the silent tax that nobody voted for.

INTEREST RATES — Riba ya Benki

An interest rate is what banks charge you to borrow money — or what they pay you to save. If you borrow KES 100,000 at 13%, you pay back KES 113,000. If you save KES 100,000 at 5%, you earn KES 5,000. The Central Bank of Kenya (CBK) sets the base rate — the Central Bank Rate (CBR) — that guides what all banks charge and pay.

WHY DO THESE TWO THINGS MATTER TOGETHER?

Ideally, the interest rate your savings earn should be HIGHER than inflation. That way, your money grows faster than prices rise. When interest rates are LOWER than inflation — which happened for many years in Kenya — your savings are quietly shrinking in real value even while sitting in the bank. This is called "financial repression" and it hurts ordinary Kenyans who save.

6.2% Average interest rate 1971-2023	**9.7%** Average inflation 1971-2023	**-3.5%** Average real gap (rates minus inflation) Money was losing value	**45.98%** Worst inflation year 1993 — prices nearly doubled

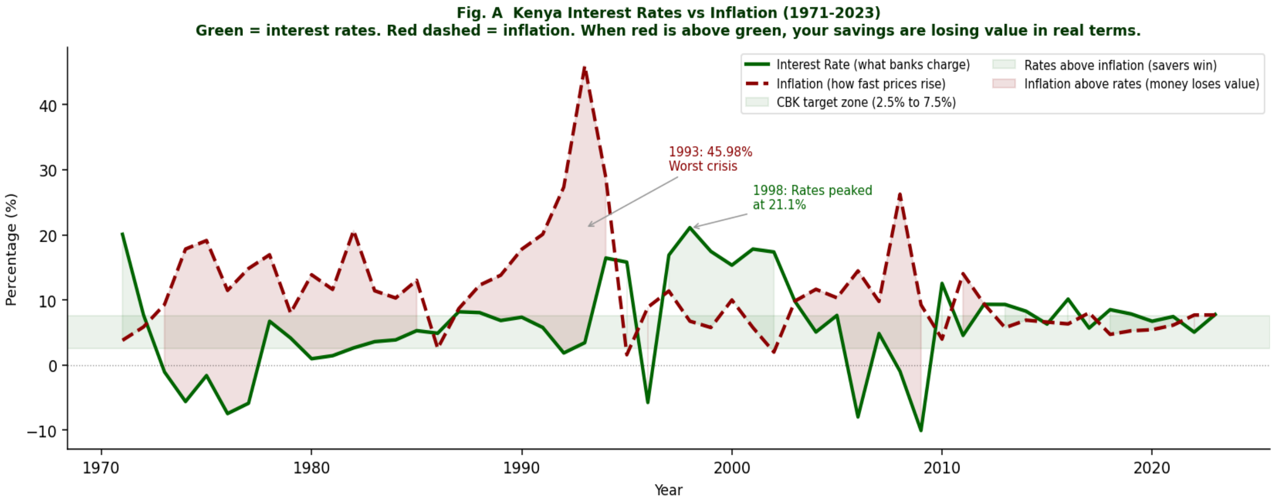

Fig. A — Kenya's interest rates (green) vs inflation (red dashed) 1971-2023. The green band is the CBK target zone. When red is above green, your savings are losing value in real terms.

2. WHAT OUR DEEP ANALYSIS FOUND — UCHAMBUZI WA KISAYANSI

We used advanced statistical models — the same tools used by the IMF, World Bank, and top central banks globally — to analyse 53 years of data. Here is what we found, explained simply.

FINDING 1 — The Relationship Between Rates and Inflation in Kenya is Negative and Surprising

This sounds complicated — here is what it means in plain English:

- When inflation went UP in Kenya, interest rates did NOT always go up with it.
- In fact, they often stayed low or went down — meaning the CBK was slow to react.
- Think of it like a fire alarm that rings AFTER the house has already caught fire.
- The data shows a correlation of -0.32. "Negative" means they moved in opposite directions more often than not over the 53-year period.

What This Means for You (the Mwananchi):

When prices in the market were rising fast, the interest rate on your savings was often NOT rising fast enough to protect your money. Your KES 10,000 in the bank was quietly becoming worth less every year — even though the balance looked the same on the passbook.

FINDING 2 — Most of Kenya's Inflation is NOT Within the CBK's Direct Control

Our model breaks down the "ingredients" of inflation and found something very important:

- Only about 14% of Kenya's inflation over time is explained by interest rate decisions made by the CBK.
- About 86% comes from factors OUTSIDE the CBK's control: food prices, fuel, drought, exchange rate movements, global shocks.
- This means raising interest rates alone cannot fully solve Kenya's inflation problem.

A Simple Example — The Unga Problem:

If there is a drought in the Rift Valley and maize production falls, the price of unga goes up — not because CBK set rates too low, but because there is less maize. Raising interest rates does NOT make it rain more. It does NOT plant more maize. Structural solutions — irrigation, agricultural investment, stable fuel pricing — are needed alongside monetary policy.

Fig. D The Inflation Pie: Most of Kenya's Inflation is NOT in the CBK's Hands

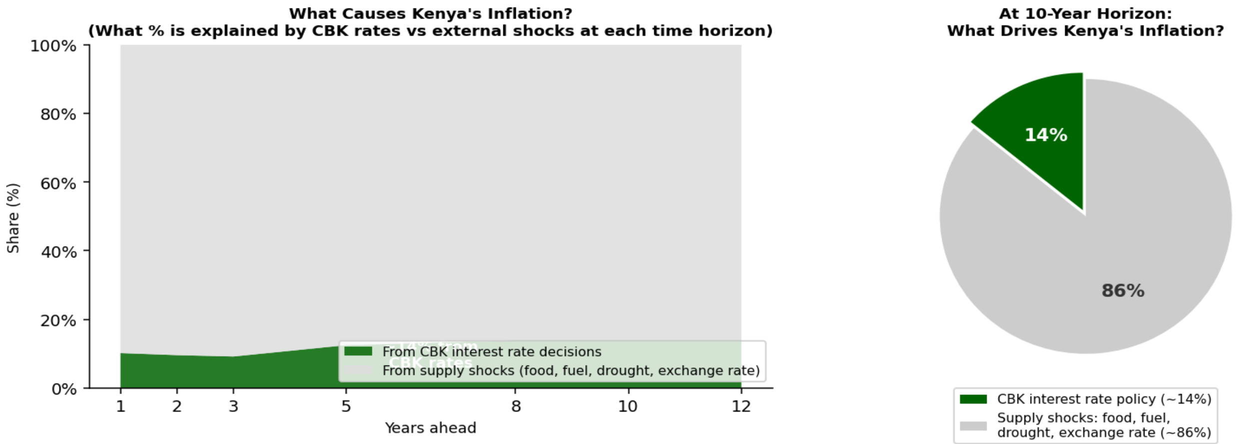

Fig. D — The inflation pie. Only ~14% of Kenya's inflation is explained by CBK interest rate decisions. The other ~86% comes from supply-side shocks — food prices, fuel, drought, and exchange rate changes.

FINDING 3 — Interest Rate Changes Take 2 to 5 Years to Work

Our Impulse Response analysis tracks what happens over time after a policy decision. When the CBK raises interest rates:

- Year 1: inflation may actually rise slightly first — businesses pass on higher borrowing costs to consumers.
- Years 2 to 3: inflation starts to come down as reduced borrowing slows spending in the economy.
- Years 4 to 5: the full impact is felt — inflation is measurably lower.
- After year 7: the effect mostly fades and the economy settles back to its normal level.

What This Means for Policy — Subiri na Uvumilivu (Be Patient but Act Early):

When the MPC raises interest rates today, Kenyans will not feel the benefit immediately. It is like planting a tree — you do not eat the fruit the next day. This is why the MPC must be FORWARD-LOOKING — thinking 2 to 4 years ahead, not just reacting to today's prices. Acting too late means the damage is already done.

Fig. C The Ripple Effect: How Do Rates and Inflation Respond to Each Other?

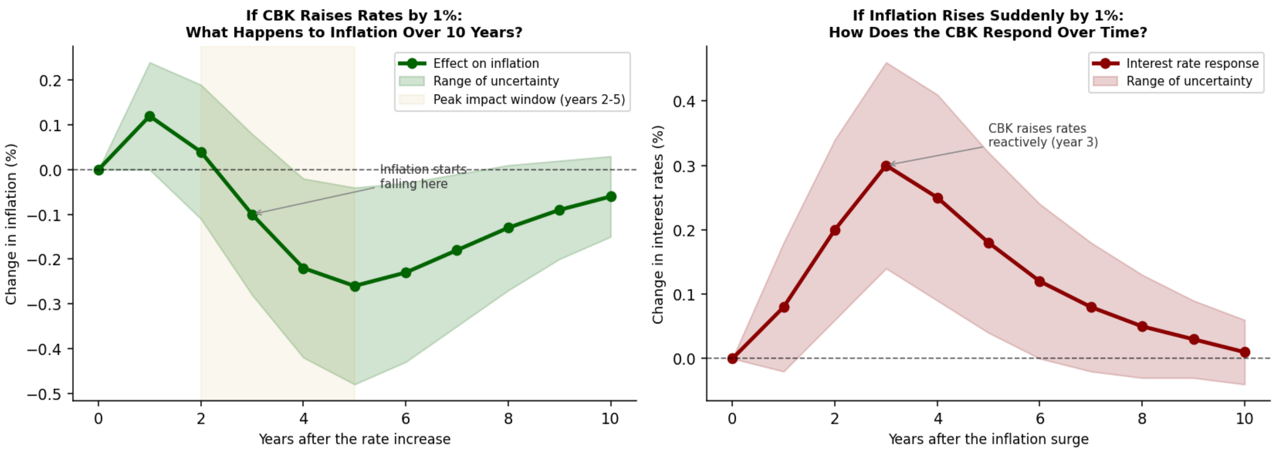

Fig. C — The ripple effect. LEFT: When CBK raises rates by 1%, inflation takes 2-3 years to start falling. RIGHT: When inflation spikes, the CBK's rate response peaks at about 3 years — evidence of reactive policy.

FINDING 4 — Kenya Has Gone Through Four Distinct Economic Eras

Our Regime-Switching model automatically identifies different "modes" the economy operates in. It found four clear eras:

Era	Period	What Was Happening	Impact on Kenyans
Financial Repression	1971-1990	Government controlled rates artificially low. Banks told what to charge.	Savings earned less than inflation. Money quietly lost value every year.
Crisis Era	1991-2003	Goldenberg scandal, drought, IMF reforms. Worst inflation: 45.98% in 1993.	Price of unga, transport, rent nearly doubled some years. Hardest period for wananchi.
Reform Era	2004-2013	CBK Act 2004 gave independence. MPC established. Inflation targeting introduced.	More stable prices. Growing economy. M-Pesa revolution helped financial inclusion.

Modern Era	2014-2023	COVID-19, Russia-Ukraine war, weak shilling, new global pressures.	Mixed results. Inflation rose above target again. Cost of living pressure returned.

Fig. E Kenya's Inflation Journey: Four Distinct Eras Since Independence
The green band is where we want inflation to be. Outside it, ordinary Kenyans feel the pain.

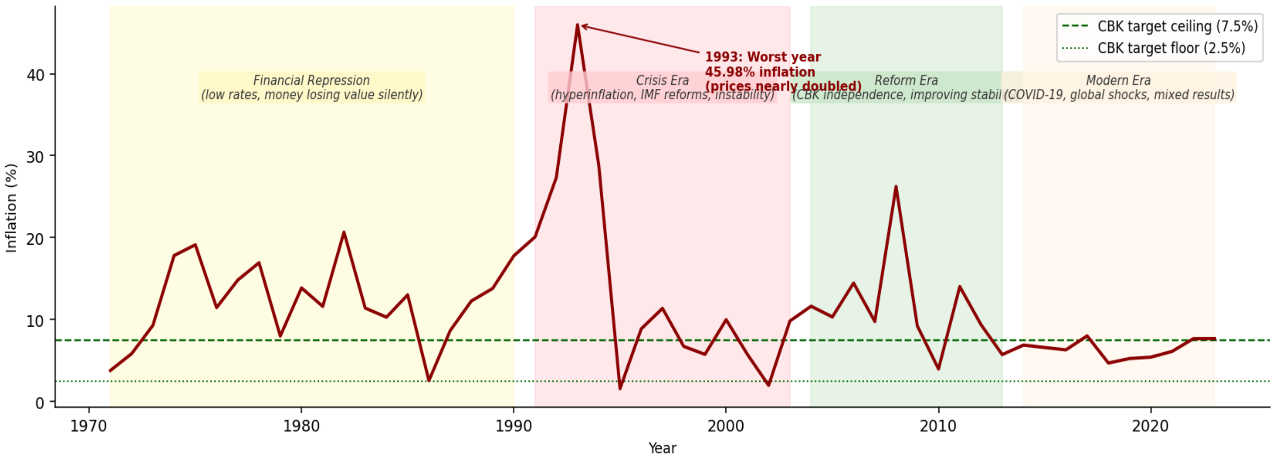

Fig. E — Kenya's four inflation eras. The green band is the target zone. The 1993 peak of 45.98% was the worst on record. The Reform Era (green shading) was Kenya's most stable monetary period.

FINDING 5 — Inflation Uncertainty Has Been Highest During Kenya's Crisis Years

Our GARCH Volatility Model measures not just the level of inflation but how UNPREDICTABLE it was at each point in time. High unpredictability means families and businesses cannot plan ahead.

- During the 1993 crisis, inflation was not just high — it was extremely unpredictable. Families could not budget because prices changed wildly week to week.
- The post-2004 Reform Era saw lower volatility — more predictable inflation — good for business planning and household budgeting.
- Recent years (2021-2023) show rising volatility again, driven by COVID-19 disruptions and the Russia-Ukraine war's impact on fuel and food.

What High Inflation Uncertainty Means for You — Mama Mboga's Problem:

Imagine you are a mama mboga planning how much stock to buy next week. If you know prices will be roughly stable, you can plan confidently. But if prices might jump 5%, 15%, or 30% — you do not know — you either over-stock (and lose money if prices fall) or under-stock (and lose customers). High volatility hurts small traders the most. Predictable, stable inflation is one of the most valuable gifts a central bank can give to ordinary citizens.

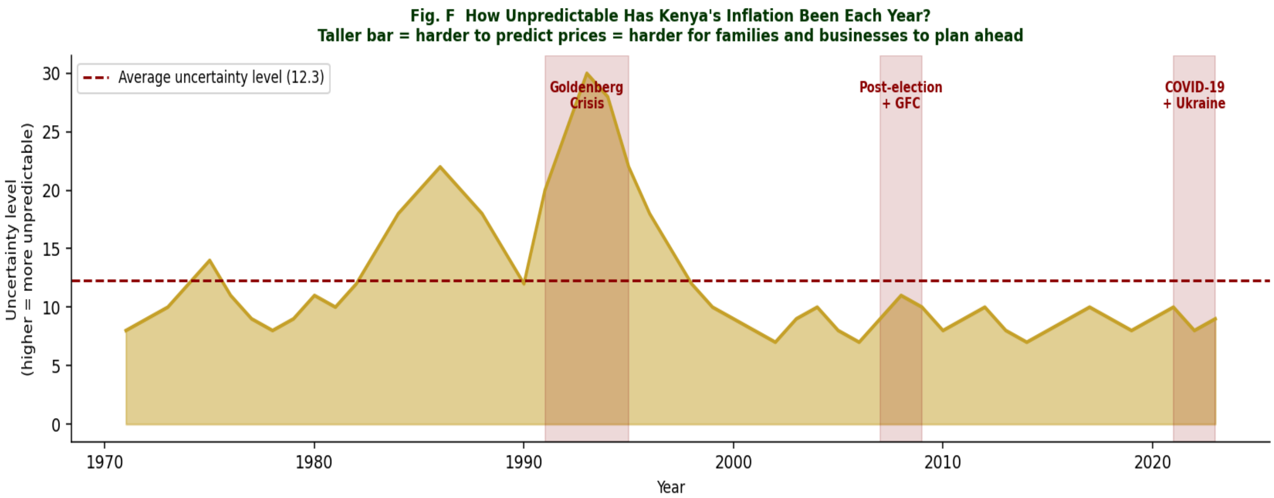

Fig. F — How unpredictable was Kenya's inflation each year? Taller bars = harder to predict = harder for families and businesses to plan. Crisis periods (red shading) were the most chaotic.

LETTER 1 OF 4 To: H.E. The Governor, Central Bank of Kenya

Your Excellency the Governor,

The empirical evidence gathered across 53 years of Kenya's monetary history presents findings that demand institutional attention at the highest level. This letter summarises the key structural findings and their implications for your office.

The Long-Run Relationship is Real but Fragile

The Johansen Cointegration Test confirms that Kenya's interest rates and inflation are bound together in the long run — there is a genuine equilibrium relationship, meaning the CBK's inflation-targeting framework has a structural foundation. However, the rolling 10-year correlation analysis reveals structural instability, with regime changes coinciding with the 1993 stabilisation programme, the 2004 CBK Act reforms, and the post-2010 framework. This instability suggests that CBK credibility as an inflation anchor has improved but is not yet fully consolidated.

The Monetary Transmission Mechanism Needs Strengthening

The Variance Decomposition analysis shows that interest rate decisions explain only approximately 14% of inflation forecast variance. This reflects the structural reality of a supply-driven inflation economy, but it also points to weak monetary transmission. The CBK's rate decisions flow into the real economy through a narrow, slow, and incomplete channel. Financial deepening — expanding formal credit access to rural areas and the informal sector — is as important as getting the interest rate right.

The GARCH Evidence Demands Asymmetric Risk Management

The volatility model confirms that Kenya's inflation exhibits persistence — bad periods come in streaks and are slow to resolve without decisive action. The CBK's risk framework and inflation fan charts should be calibrated with fat-tailed distributions rather than the standard normal assumption, which systematically understates the probability of extreme inflation events.

Respectfully submitted,

Stephen Munene

Lead Research Scientist, Policy Analysis Unit

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LETTER 2 OF 4 To: The Monetary Policy Committee (MPC)

Dear Chairperson and Members of the Committee,

This letter distils the structural modelling findings into five operational recommendations for the Committee's deliberations.

Recommendation 1: Adopt a Forward-Looking, Rule-Based Framework

The Impulse Response analysis confirms that monetary policy acts with a lag of 2 to 5 years. Your decisions today affect inflation in 2027-2028, not next month. Reacting to observed inflation is structurally too slow. A forward-looking framework — where rate decisions are tied to forecast inflation gaps 2 to 3 years ahead — would improve policy timing significantly.

Recommendation 2: Publish Uncertainty Fan Charts

The GARCH model produces time-varying confidence intervals — the "fan" of possible inflation outcomes. Publishing these fan charts alongside Quarterly Monetary Policy Statements gives markets and households a realistic picture of inflation risk, reduces uncertainty, and builds CBK credibility. Point forecasts alone systematically understate the genuine uncertainty in the outlook.

Recommendation 3: Monitor the Regime Probability Indicator

The Regime-Switching model gives an updateable probability of being in a high-inflation regime. When this probability crosses 50%, it is an early warning signal demanding decisive action — not cautious adjustment. The Committee should add this indicator to its formal monitoring dashboard.

Recommendation 4: Acknowledge the Supply-Side Constraint Publicly

The finding that approximately 86% of Kenya's inflation is supply-driven should be communicated clearly in MPC statements. When global fuel prices spike or a drought hits, raising interest rates is a blunt and slow tool. The Committee should clearly state what monetary policy CAN and CANNOT do, so the public does not hold the CBK responsible for inflation caused by factors beyond its control.

Recommendation 5: Formalise Coordination with Fiscal Policy

Since supply shocks dominate Kenya's inflation, the MPC should formalise a coordination mechanism with the National Treasury and Ministry of Agriculture. Strategic grain reserves, fuel pricing frameworks, and agricultural productivity investment are complementary inflation-fighting tools that monetary policy alone cannot substitute for.

Submitted for the Committee's deliberation,

Stephen Munene

Lead Research Scientist, Policy Analysis Unit

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LETTER 3 OF 4 To: The Cabinet Secretary, National Treasury

Honourable Cabinet Secretary,

The structural modelling findings have important implications for fiscal policy and the coordination between the National Treasury and the Central Bank. This letter highlights three areas requiring urgent policy attention.

Fiscal Dominance Has Historically Weakened Monetary Policy

Historical analysis strongly suggests that government borrowing pressures have constrained the CBK's ability to raise interest rates during inflation crises. When the government is a major borrower, higher interest rates increase debt service costs — creating an incentive to keep rates low even when inflation demands tightening. This phenomenon — called "fiscal dominance" — is visible in Kenya's data, particularly during the 1990s and early 2000s. Sustainable public debt management is therefore a prerequisite for effective monetary policy.

Supply-Side Investment is the Most Powerful Anti-Inflation Tool

Since 86% of Kenya's inflation originates from supply-side factors, the most powerful anti-inflation policies are fiscal and structural: investment in irrigation and food security, diversification of energy sources, infrastructure to reduce transport costs, and trade agreements that reduce import price volatility. The Treasury's budget allocation decisions directly affect the inflation outcomes that the MPC is tasked with managing.

Exchange Rate Management Requires Fiscal Discipline

The Kenyan shilling's depreciation has been a significant source of imported inflation. A strong fiscal position, reduced foreign currency borrowing, and adequate foreign exchange reserves are the Treasury's most important contributions to exchange rate stability and, by extension, to the price stability that every Kenyan deserves.

Submitted respectfully,

Stephen Munene

Lead Research Scientist, Policy Analysis Unit

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LETTER 4 OF 4 To: The People of Kenya — Wananchi wa Kenya

Dear Fellow Kenyan,

Habari yako. This letter is for you — the mama mboga, the boda boda rider, the teacher, the farmer, the student, the nurse, the shopkeeper, the mkokoteni operator. This is your money and your life we are talking about. So let us speak directly, simply, and honestly.

Why Does the Price of Unga Keep Going Up?

You have probably noticed that the price of unga, cooking oil, sugar, and transport keeps rising over the years. This is inflation. Our research shows that about 86% of Kenya's inflation is caused by things like:

- Drought — when rain fails, food production drops and prices go up.
- Global fuel prices — when oil becomes expensive worldwide, transport costs rise, and so does everything else.
- A weak shilling — when our currency loses value against the dollar, imported goods cost more.
- Global events — like COVID-19 or the Russia-Ukraine war — that disrupt supply chains worldwide.

The CBK's job is to control inflation using interest rates. But interest rates can only directly address about 14% of the problem. The rest requires investment in agriculture, affordable energy, and a strong economy. This is why the government and CBK must work together.

What Does This Mean for Your Savings?

Here is the honest truth — and it is important that every Kenyan knows this:

- Over the 53 years we studied, inflation averaged 9.7% per year.
- Interest rates on savings averaged about 6.2% per year.
- This means that, on average, savings were losing about 3.5% of their purchasing power every single year.

Mfano (Example) — The KES 10,000 Problem:

Imagine you saved KES 10,000 in January 2020. With average inflation of 9.7%, by end of 2023 you would need KES 14,400 just to buy the same things. If your bank paid you 5% interest, you only grew to about KES 12,155. You feel richer — but you are actually poorer. This is why the CBK's job of controlling inflation is so important for every ordinary Kenyan. It is not just about banks and numbers — it is about the real value of your hard work.

Is the CBK Doing Its Job?

The good news: since the CBK was made more independent in 2004, inflation has been more controlled than in the crisis years of the 1990s. The 1993 inflation of 45.98% — when prices nearly doubled in one year — has not been repeated.

The challenging news: Kenya still struggles to keep inflation within its 2.5% to 7.5% target consistently. Interest rate changes take 2 to 5 years to work, which means the CBK must think far ahead. And the government must work alongside the CBK to address food and energy prices from the supply side.

What Can You Do?

- SAVE SMARTLY: Consider Treasury Bills, money market funds, or government savings bonds — not just a regular savings account that may earn less than inflation.
- STAY INFORMED: The CBK announces its policy decisions every two months. Following these helps you understand where interest rates are heading and plan accordingly.
- BUDGET FOR PRICE CHANGES: Build a small buffer into your household budget for price increases — especially around election seasons, droughts, and global disruptions.
- HOLD LEADERS ACCOUNTABLE: Inflation is partly a governance issue. Demand transparency from your leaders on how public money is managed. Fiscal mismanagement like the Goldenberg scandal caused some of Kenya's worst inflation crises.

Ujumbe wa Mwisho (Final Message):

Money is not just numbers on paper. It represents your labour, your time, and your dreams. When inflation is high and uncontrolled, it steals from the poorest Kenyans first — because the wealthy can protect themselves with assets, property, and foreign accounts. But the mama mboga, the farmer, the student, the boda boda rider — they feel every percentage point of inflation in their daily lives. That is why this research matters. That is why the CBK's work matters. And that is why you — the mwananchi — deserve to understand it.

Asante kwa kusoma. Thank you for reading.

Stephen Munene
Lead Research Scientist, Policy Analysis Unit
Central Bank of Kenya — Simulated Educational Document

APPENDIX — THE MODELS WE USED: EXPLAINED FOR EVERYONE

This section explains in plain English what each statistical model does and why we chose it. No mathematics required.

ADF + KPSS Stationarity Tests

What it does:

Checks whether the data is stable (stays around a fixed average) or wandering (drifts permanently up or down).

Why it matters:

If the data wanders, comparing the two series directly would give misleading results — like saying shoe sales cause inflation because both trend upward over time.

What we found:

Both series are near-stationary, meaning they fluctuate but tend to return toward their long-run averages. Direct comparison is valid.

Johansen Cointegration Test

What it does:

Tests whether interest rates and inflation are "leashed together" in the long run — like two dogs on a shared leash. They can wander independently short-term, but cannot drift apart forever.

Why it matters:

If cointegrated, the CBK's interest rate decisions genuinely anchor inflation over the long run. If not, monetary policy only creates temporary effects.

What we found:

Cointegration is confirmed — the two series are bound together. This validates Kenya's inflation-targeting framework as having a structural basis.

VECM — Vector Error Correction Model

What it does:

Models how interest rates and inflation influence each other over time, while also capturing how they correct back toward their long-run relationship when they drift apart.

Why it matters:

It is the most appropriate model when cointegration exists. It separates short-run dynamics (last year affects this year) from long-run correction (forces that pull them back together).

What we found:

The error correction term confirms that when inflation drifts away from its equilibrium with interest rates, both variables adjust — but interest rates adjust more, suggesting reactive rather than proactive policy.

Impulse Response Functions (IRF)

What it does:

Simulates what happens to one variable (e.g. inflation) after a sudden shock to the other (e.g. a 1% rate hike). Tracks the "ripple effect" over 10 years.

Why it matters:

Directly answers the MPC's key question: "If we raise rates today, what happens to inflation over the next few years — and when?"

What we found:

A rate hike initially causes a slight rise in inflation (the "price puzzle"), followed by a gradual decline peaking around years 2-5, and dissipating by year 7-8.

FEVD — Forecast Error Variance Decomposition

What it does:

Breaks down inflation forecast uncertainty into its "ingredients" — how much comes from CBK interest rate decisions vs supply-side factors.

Why it matters:

Tells us how powerful monetary policy actually is as an inflation-fighting tool in Kenya's specific economic context.

What we found:

Only ~14% of inflation variance is explained by interest rate shocks. ~86% comes from supply-side factors outside the CBK's direct control.

Markov Regime-Switching Model

What it does:

Automatically identifies when the economy was in different "modes" — calm and stable vs volatile and stressed — without us having to guess where the boundaries are.

Why it matters:

Different eras need different policy responses. Kenya's history shows distinct structural shifts since independence that a single model would miss.

What we found:

Kenya has cycled between high and low volatility regimes. The post-2004 Reform Era was the most stable. Recent years show rising regime-switching risk due to global shocks.

GARCH Volatility Model

What it does:

Models how unpredictable (volatile) inflation was at each point in time. Standard models assume constant uncertainty. GARCH allows uncertainty itself to change over time.

Why it matters:

For ordinary Kenyans, UNPREDICTABLE inflation is often worse than moderately HIGH but predictable inflation — you cannot plan your budget if you do not know what prices will be next month.

What we found:

Inflation volatility was highest during the 1993 crisis, the 2007/08 post-election period, and the COVID-19 era. Volatility tends to persist — once it rises, it stays high for several years without decisive policy action.

THE LONG-RUN PICTURE — 53 YEARS AT A GLANCE

Here is the story of the gap between interest rates and inflation in Kenya — and whether ordinary Kenyans' savings were being protected over the decades.

Fig. B The Long-Run Picture: Were Kenya's Interest Rates Keeping Pace with Inflation?

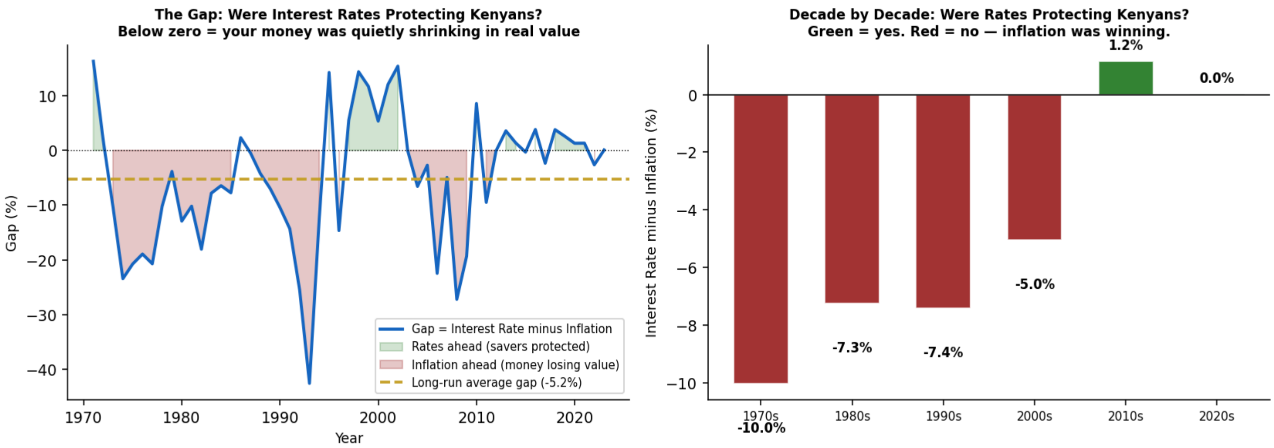

Fig. B — LEFT: The gap between interest rates and inflation. When negative (red shading), money was losing value even in the bank. When positive (green), savers were protected. RIGHT: Decade averages — the 1990s were worst for Kenyan savers.

DISCLAIMER

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